Supplementary material for:

What genome-scale Perturb-seq responses contain: an independent-screen analysis and the limits of reading perturbation responses as regulatory networks

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References to Sections, Figures and Tables without an "S" prefix refer to the main text.

The analyses below started the project. They concern specific estimators, simulators and regimes, and their conclusions are stated at that level. All simulation results use the seeds listed. "Diagnostic re-implementations" of published method families (PHOENIX-, DCDI-, CellOracle- and GENIE3-style) implement only each method's core idea in our code. They are controls, not evaluations of the original software.

Metric definitions (all simulations). The candidate-edge universe is every ordered pair $(j \to i)$ of distinct *measured* genes: $70 \times 69 = 4{,}830$ pairs in the 80-gene simulator, of which about 114 are true edges. AUROC and average precision (AUPRC) are computed over this universe. AUPRC× is AUPRC divided by the base rate (true edges / candidates), so 1 = random. E is the number of true edges among measured genes; "top-E" metrics use the E highest-scored pairs. "Indirect" pairs have a directed path of length ≥ 2 (through any gene, measured or hidden) and no direct edge.

S1. Thermodynamic occupancy model

Occupancy calibration. For each TF, E_{thr} is the 10^{-4} quantile of PWM energies on 400,000 background positions with promoter base composition (a FIMO-like threshold). Occupancy uses $\mu = E_{\text{thr}} + \{-4, -2, 0\}$ (-2 pre-registered as default). ProBound models (MotifCentral v1.0.0; human model preferred, lowest fit id) were rescaled to a maximum log-affinity of 0.

T1, synthetic genome (1,500 genes, 20 JASPAR TFs, 3 seeds). Two different comparisons are reported.

- Over all targets, summed occupancy at $\mu = E_{\text{thr}}$ reached AUROC 0.959 vs 0.820 for the maximum motif score.
- Restricted to weak-cluster targets vs non-targets, occupancy reached 0.968 vs 0.647.

A fitted occupancy model recovered TF activities μ (Spearman 0.976).

T3, cell-state μ (K562 vs HepG2). Test R^2 for log expression: 0.172 (accessibility only), 0.185 (fixed μ), 0.188 (state-specific μ). For the HepG2−K562 *difference*, R^2 rose from 0.006 (fixed μ) to 0.017 (state-specific μ), an improvement of 0.011. The inferred $\Delta\mu$ did not correlate with independent proxies (Spearman 0.06–0.09, permutation $p > 0.29$).

ProBound vs JASPAR (122 TFs). Binding AUROC was higher with ProBound for 69–73% of TFs (paired Wilcoxon $p < 10^{-7}$), by about 0.02. There was no gain for function (13 TFs).

S2. Latent neural ODE on a three-gene cascade, and a 10-gene toy

Simulation. A→B→C with Hill kinetics ($K = 0.6$, $n = 2$), Euler–Maruyama ($dt = 0.005$). 200 cells (150 train), 41 observations at interval 0.2, measurement noise 5% or 20% of the per-gene s.d. **Model.** Gene-aligned latent states (residual per-gene MLP encoder and decoder) and dynamics $dz/dt = W\sigma(a(z-c)) - \text{softplus}(g)z + b$ with $\text{diag}(W) = 0$. Free latent units are initialised from a learned constant ("from start") or from $(x_t, x_t - x_{t-1})$. Loss: K-step rollout MSE (4 Euler substeps per interval) + reconstruction + $\lambda\|W\|_1$. Adam, learning rate 10^{-2}, cosine schedule, 2,000 iterations (1,200 for from-start runs). λ grid $\{0, 10^{-5}, \ldots, 10^{-2}\}$; 3 seeds. An edge is called if its strength ($|W_{ij}| \times$ range of j's activation along the rollout) is $\geq 10\%$ of the largest. **Results.** With all genes observed, the exact graph was recovered in 57–71% of settings. With B hidden, the shortcut A→C was called in 81–100% of fits. With knockdowns of the hidden B supplied and modelled (applied to a latent unit with a learned efficiency), A→ h →C was recovered without the shortcut in 75% of

settings (latent–B correlation 0.975). With the same knockdowns unmodelled, the shortcut was always called. In a 10-gene toy, a binding prior raised AUROC from 0.749 to 0.947 (0.962 with knockdown-aware fitting). Two indirect edges stayed in the top 9 until the intermediate TF was itself knocked down.

S3. 80-gene simulator: ablations and experiment selection

Simulator (in-house, SERGIO-style): 80 genes (16 TFs, 8 non-TF regulators, 56 targets; about 140 edges; TF cascades of depth ≥ 3; 70% activators). Per-edge Hill coefficient $n \in \{1..4\}$ and threshold $K \sim U(0.5, 2)$; decay $U(0.5, 1)$; process noise 0.05; a hidden cell-size factor; Poisson counts with log-normal library size; noisy labelling velocity ($1\times$ s.d.); 10 unmeasured genes; 1,500 wild-type cells and 80 cells per knockdown, with efficiency $U(0.05, 0.3)$. Seeds 0–9. **BOND-V:** nodewise adaptive lasso [1] (BIC) of velocity on Hill features $\phi(x_j) = x_j^2/(K_j^2 + x_j^2)$, $K_j = \text{median}(x_j)$, plus a decay term. Features are standardised before division by the penalty factor $1 - \alpha p_{ij}$. Each gene's own-knockdown cells are dropped from its equation. Ablation results (AUPRC×, 10 seeds):

- full model 18.6; without prior 15.5; ignoring interventions 17.6; static (no velocity) 16.4;
- knockdown mean-difference 26.4, but with 36% indirect edges in its top E;
- diagnostic re-implementations: CellOracle-style 8.9, GENIE3-style [2] 5.7.

Disambiguation-driven knockdown selection beat random and hub-based choice in 10/10 seeds (area under the AUPRC× coverage curve $+0.95 \pm 0.67$ vs random). It was not tested on real data.

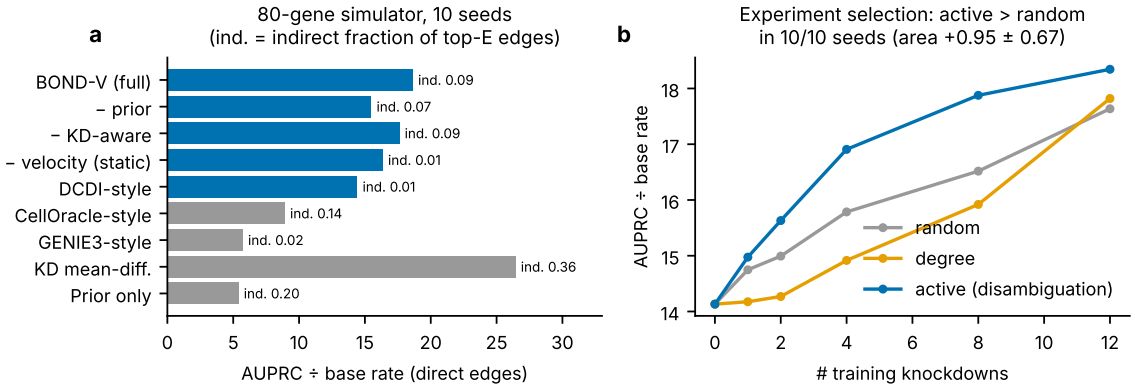


Figure S1: 80-gene simulator. (a) Ablations and diagnostic re-implementations. (b) Knockdown selection strategies.

S4. Linear SEM on real held-out knockdowns

On 203 strongly knocked-down Replogle genes, a knockdown-aware steady-state linear SEM predicted held-out knockdown profiles with correlation 0.012 ± 0.298, against 0.422 ± 0.402 for the mean training profile. ChIP and motif priors did not change this.

S5. A lag-feature ridge detector of hidden regulators (memory)

Scope. This tests one estimator: ridge regression of an observed gene's 5-step-ahead change on the current observed state, with and without lags of 1, 2, 4 and 8 samples. It is not a test of memory-function methods in general [3–5]. **Simulation.** Measurement-noise levels are relative to each gene's simulated within-trajectory fluctuation s.d. They are not calibrated against any real assay, so we do not call any level "realistic". **Results.** With regulator process noise four times that of other genes and clean data, the detector ranked children of hidden regulators above other genes (AUROC 0.84 vs 0.51 with nothing hidden; Table S1). It failed with weaker regulator fluctuations, with measurement noise $\geq 0.5\times$ the fluctuation s.d., and with sampling intervals ≥ 0.25. Estimates of the hidden regulator's timescale and of the number of hidden regulators (lagged-residual rank) failed.

Table S1: Lag-feature ridge detector (AUROC; 80-gene time courses; seeds 0–9 or 0–5).

Detector / condition	AUROC (mean ± sd over seeds)	seeds
Memory gain, 2 hidden regulators (clean)	0.836 ± 0.169	10
Same genes, nothing hidden (control)	0.512 ± 0.114	10
Snapshot detector ($1-R^2$), hidden regulators	0.374 ± 0.068	10
Memory gain, weak regulator fluctuations	0.543 ± 0.127	10
Memory gain, measurement noise 0.1× fluct. sd	0.875 ± 0.129	6
Memory gain, measurement noise 0.25× fluct. sd	0.705 ± 0.151	6
Memory gain, measurement noise 0.5× fluct. sd	0.487 ± 0.052	6
Memory gain, measurement noise 1.0× fluct. sd	0.444 ± 0.058	6
control (nothing hidden), noise 0.25	0.545 ± 0.103	6
control (nothing hidden), noise 0.5	0.535 ± 0.050	6
Noise 0.5 + causal smoothing (attempted fix)	0.531 ± 0.105	6
Sampling interval 0.05 (clean)	0.807 ± 0.072	4
Sampling interval 0.1 (clean)	0.881 ± 0.072	4
Sampling interval 0.25 (clean)	0.664 ± 0.138	4
Sampling interval 0.5 (clean)	0.463 ± 0.253	4
Sampling interval 1.0 (clean)	0.436 ± 0.024	4
Sampling interval 2.0 (clean)	0.433 ± 0.037	4

Median memory gain (fraction of held-out error removed by the past), clean main condition: genes with a hidden regulator 0.027 ± 0.023, other genes -0.007 ± 0.002. Oracle gain from adding the true hidden state: 0.093 ± 0.054 vs -0.001 ± 0.000.

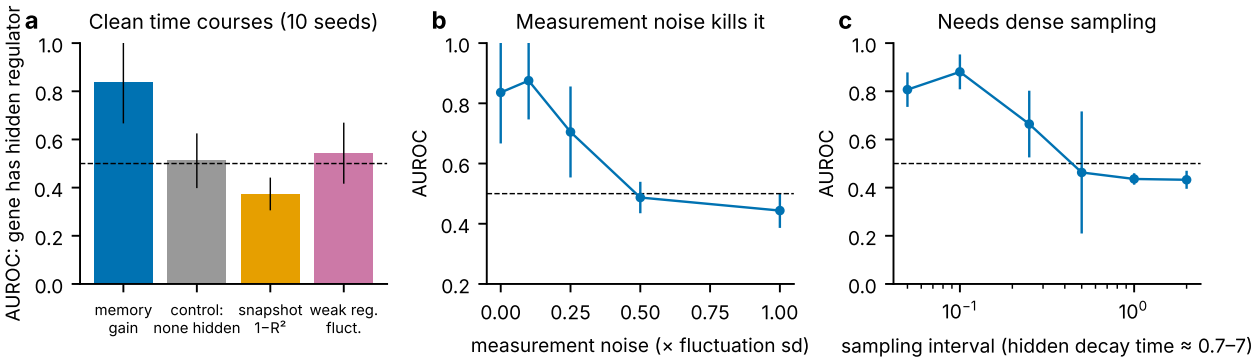


Figure S2: Lag-feature ridge detector. (a) Clean, dense time courses. (b) Measurement noise relative to simulated fluctuations. (c) Sampling interval.

S6. Intervention-induced indirect edges (errors-in-variables)

For A→B→C with $B = bA + \varepsilon_B$, $C = cB + \varepsilon_C$ and B measured as $\tilde{B} = B + e$, regressing C on $(A, \tilde{B})$ gives $\beta_{A\to C} = cb\lambda$, $\lambda = \sigma_e^2/(\sigma_e^2 + \mathrm{Var}(B|A))$ [6]. Perturbing A leaves β unchanged but makes it easier to detect. Perturbing B lowers λ (Table S2). In the 80-gene simulator, perturbing the source raised the indirect-call rate only slightly, and hidden mediators dominated (Table S3). A simple identifiability flag was modestly informative (Table S4).

S7. Prior forms and imperfect regulator masks

With an *exact* regulator list, a hard identity mask raised AUPRC× from 15.5 (no prior) to 22.2, and adding a soft site-level binding prior on top gave 23.2 (Table S5). The reading is "mask first; a site prior adds a little on top". A hard binding-site mask capped recall at 25% and was harmful. With imperfect masks (Table S6, Fig. S3), each missing regulator cost far more than each false one, because a missing regulator's true edges cannot be recovered at all.

- Dropping 25% of regulators (6 of 22) brought AUPRC× down to the no-prior level (15.9 vs 15.5); dropping 50% fell well below it (11.2).
- Adding false regulators degraded performance gradually: 19.1 with 50% extra, and 15.9 only with 200% extra (44 false regulators).

Table S2: Three-gene check ($b = c = 1$, $\sigma_B = \sigma_C = \sigma_e = 0.5$, $n = 2{,}000$, 300 replicates).

Condition	theory $cb\lambda$	simulated coefficient	mean t	Lasso selects A→C
observational	0.500	0.497 ± 0.024	21.0	1.00
perturb A	0.500	0.500 ± 0.022	23.9	1.00
perturb B	0.100	0.099 ± 0.017	5.5	1.00
perturb A and B	0.100	0.076 ± 0.011	6.8	1.00

Table S3: Indirect pairs called as edges by perturbation status (BOND-V without prior; seeds 0–9 × 3 random halves of the perturbation set).

Source (regulator)	2-hop mediator	indirect pairs	% called (top-E), ± seed sd
source perturbed	mediator unperturbed	874	5.9 ± 3.2
source perturbed	mediator perturbed	881	4.9 ± 3.4
source perturbed	mediator hidden	175	14.9 ± 14.3
source perturbed	longer path only	1709	0.2 ± 0.3
source unperturbed	mediator unperturbed	779	5.1 ± 1.9
source unperturbed	mediator perturbed	862	4.3 ± 3.3
source unperturbed	mediator hidden	179	17.3 ± 12.5
source unperturbed	longer path only	1723	0.3 ± 0.5

- A mask with both errors (25% missing, 100% false) was worse than no prior (10.7). Adding the soft site prior recovered only part of the loss (12.4).

A hard mask is therefore worthwhile only when the regulator annotation has high recall; with lower recall, soft priors or no prior are safer.

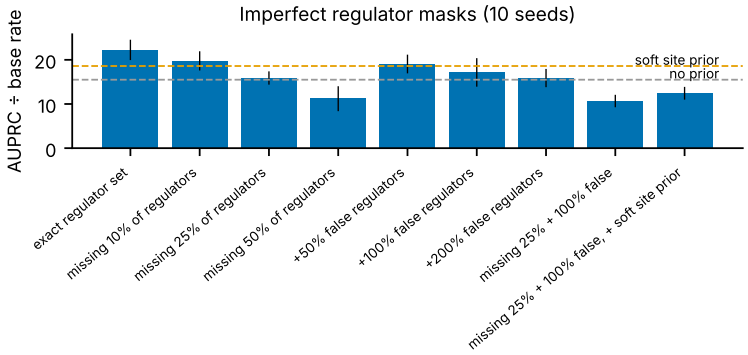


Figure S3: Imperfect regulator masks.

S8. Propagation depth: true vs estimated operator

We evaluated every regulator (seeds 0–9), stratified by its true downstream depth, and scored both operators against the *same* targets on the *same* log-normalised scale (Table S7, Fig. S4).

- **Targets:** (i) the noise-free steady-state knockdown effect, mapped to the log scale with the first-order transform of the simulator's normalisation; (ii) the measured mean effect in the simulated data.
- **True operator:** the linearised production Jacobian at the wild-type steady state, with its predictions mapped to log scale by the same transform.
- **Estimated operator:** BOND-V, 5-fold over knockdowns.

Table S4: Identifiability flag (edge $j \to i$ flagged if some unperturbed m has predicted $j \to m$ and $m \to i$ edges among the top $2E$).

Quantity	Value
n predicted edges	3426
fraction flagged	0.437
precision flagged	0.273
precision unflagged	0.441
indirect frac flagged	0.089
indirect frac unflagged	0.054
share of indirect FPs that are flagged	0.561
share of TPs flagged	0.324
precision source perturbed	0.835
precision source unperturbed	0.208
precision regulator source perturbed	0.835
precision regulator source unperturbed	0.717
n edges from unperturbed regulators	739
n edges from perturbed	873
fraction of edges from non regulators	0.529
precision flagged among regulator sources	0.711
precision unflagged among regulator sources	0.820

Table S5: Prior forms (exact regulator list; seeds 0–9; mean $\pm$ s.d.).

Prior form	recall ceiling	AUROC	AUPRC×	Precision@E	indirect frac.	held-out KD corr
no prior	1.00	0.834 ± 0.021	15.483 ± 2.246	0.382 ± 0.033	0.069 ± 0.026	0.332 ± 0.139
soft site prior (a=0.8)	1.00	0.872 ± 0.018	18.638 ± 2.361	0.467 ± 0.021	0.093 ± 0.024	0.418 ± 0.163
soft identity prior (non-regulators x3 penalty)	1.00	0.906 ± 0.013	20.631 ± 2.677	0.493 ± 0.023	0.106 ± 0.036	0.416 ± 0.136
hard identity mask (TFs + other regulators)	1.00	0.916 ± 0.018	22.248 ± 2.303	0.496 ± 0.031	0.140 ± 0.052	0.394 ± 0.106
hard TF-only mask	0.80	0.835 ± 0.023	19.334 ± 2.947	0.457 ± 0.054	0.219 ± 0.068	0.232 ± 0.076
hard binding mask (TF x top-15% occupancy)	0.25	0.617 ± 0.027	7.296 ± 1.169	0.224 ± 0.016	0.188 ± 0.052	-0.001 ± 0.102
hard identity mask + soft site prior	1.00	0.924 ± 0.012	23.186 ± 2.258	0.515 ± 0.032	0.155 ± 0.058	0.397 ± 0.102

Remaining mismatches between the two operators include linearisation error, and cell-to-cell heterogeneity (input levels, cell size) that the true operator, computed at mean inputs, ignores. The noise-free and measured targets correlate 0.79 on average.

For regulators with downstream depth ≥ 3, the mean over $D \in \{2,3,4\}$ minus $D = 1$ (bootstrap 95% CI over 82 regulators) was:

- true operator: $+0.19$ [0.16, 0.23] against the noise-free target and $+0.07$ [0.05, 0.09] against the measured target;
- estimated operator, stabilised (rescaled to spectral radius 0.8): $+0.04$ [0.01, 0.07] against the noise-free target and -0.02 [-0.05, -0.00] against the measured target;
- estimated operator, unstabilised: strongly negative (-0.35 and -0.51).

Conclusion: deeper propagation helps with the true operator in this simulator, against both noise-free and measured targets. The estimated pipeline does not reproduce that benefit: there is no consistent improvement beyond one step after stabilisation, and collapse without it. At $D = 1$ on the measured target, the estimated operator matched or exceeded the true operator (0.70 vs 0.64 for deep regulators), consistent with the heterogeneity mismatch above. Table S8 reports the earlier comparison, in which the true operator was scored on noise-free raw-scale effects. In Table S9, the $(I-J)^{-1}$ column is an algebraic inverse. It equals the infinite propagation sum only when $\rho(J) < 1$, which holds only for the stabilised rows.

Table S6: Imperfect regulator masks (BOND-V; seeds 0–9). Missing regulators are dropped from the mask; false regulators are random non-regulator genes added to it.

Regulator mask	allowed regulators	recall ceiling	AUPRC×	Precision@E	held-out KD corr
exact regulator set	22	1.00	22.2 ± 2.3	0.496	0.39
missing 10% of regulators	20	0.90	19.8 ± 2.2	0.454	0.32
missing 25% of regulators	16	0.73	15.9 ± 1.5	0.405	0.22
missing 50% of regulators	11	0.50	11.2 ± 2.8	0.316	0.10
+50% false regulators	33	1.00	19.1 ± 2.1	0.433	0.40
+100% false regulators	44	1.00	17.1 ± 3.2	0.429	0.41
+200% false regulators	66	1.00	15.9 ± 2.1	0.395	0.37
missing 25% + 100% false	38	0.72	10.7 ± 1.4	0.301	0.31
missing 25% + 100% false, + soft site prior	38	0.72	12.4 ± 1.4	0.336	0.29
Reference: no prior	–	1.00	15.5 ± 2.2	0.382	0.33
Reference: soft site prior only	–	1.00	18.6 ± 2.4	0.467	0.42

S9. Context: earlier sign-flip summary and Shifrut 2018

Table S10 gives a sign-flip summary that conditions on targets strong in both contexts; Table 7 gives the unconditional, noise-matched analysis.

S10. Additional checks for the main-text analyses

S11. Reproducibility

Environments: main-text and Supplement S5–S10 analyses were run with Python 3.13 and the versions in `requirements.txt`; Supplement S1–S4 (including the PyTorch latent-ODE runs) were run mainly on a second machine with Python 3.10.12 and the versions in `requirements-s1-s4.txt` (torch 2.14.1+cpu); some 80-gene coverage runs used the main environment. `run_all.sh` lists the commands used, in order, starting from the raw files listed in `data/README.md`. End-to-end regeneration from a clean environment was verified only for the analyses stated in the main-text Methods; the other commands are supplied but not re-verified. Random seeds are fixed in each script (`numpy.random.default_rng` with the seeds stated above). Approximate run times on a 2-core machine: decomposition and benchmark $<$10 min; cascade test with positive control about 10 min; 80-gene simulations 1–3 h.

References

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Table S7: Propagation depth, both operators on the same targets and scale (correlation with the knockdown effect; mean over regulators; seeds 0–9).

Operator	target	regulators (n)	D=1	2	3	4	6	8
true operator	noise-free	depth 1 (95)	0.91	0.91	0.91	0.91	0.91	0.91
true operator	noise-free	depth 2 (43)	0.78	0.88	0.88	0.88	0.88	0.88
true operator	noise-free	depth $\geq$ 3 (82)	0.57	0.73	0.77	0.78	0.78	0.78
true operator	measured	depth 1 (95)	0.71	0.71	0.71	0.71	0.71	0.71
true operator	measured	depth 2 (43)	0.71	0.75	0.75	0.75	0.75	0.75
true operator	measured	depth $\geq$ 3 (82)	0.64	0.70	0.71	0.71	0.71	0.71
estimated	noise-free	depth 1 (95)	0.39	0.18	0.06	0.04	0.01	-0.00
estimated	noise-free	depth 2 (43)	0.55	0.38	0.08	-0.00	0.05	-0.03
estimated	noise-free	depth $\geq$ 3 (82)	0.57	0.54	0.20	-0.06	0.06	-0.04
estimated	measured	depth 1 (95)	0.36	0.16	0.05	0.02	0.00	-0.01
estimated	measured	depth 2 (43)	0.58	0.37	0.07	-0.01	0.06	-0.06
estimated	measured	depth $\geq$ 3 (82)	0.70	0.54	0.12	-0.10	0.10	-0.05
estimated, stabilised	noise-free	depth 1 (95)	0.39	0.33	0.31	0.33	0.33	0.33
estimated, stabilised	noise-free	depth 2 (43)	0.55	0.53	0.50	0.50	0.51	0.51
estimated, stabilised	noise-free	depth $\geq$ 3 (82)	0.57	0.63	0.62	0.61	0.61	0.61
estimated, stabilised	measured	depth 1 (95)	0.36	0.30	0.29	0.30	0.30	0.30
estimated, stabilised	measured	depth 2 (43)	0.58	0.55	0.51	0.51	0.52	0.52
estimated, stabilised	measured	depth $\geq$ 3 (82)	0.70	0.71	0.67	0.66	0.68	0.68

Table S8: Propagation depth, earlier comparison on different targets (true operator: noise-free raw-scale effects; estimated: measured log-scale effects). Superseded by Table S7.

Operator	held-out regulators (n)	D=1	2	3	4	6	8
true operator	downstream depth 1 (95)	0.90	0.90	0.90	0.90	0.90	0.90
true operator	depth 2 (43)	0.76	0.85	0.85	0.85	0.85	0.85
true operator	depth $\geq$ 3 (82)	0.61	0.74	0.78	0.78	0.78	0.78
estimated	downstream depth 1 (95)	0.36	0.16	0.05	0.02	0.00	-0.01
estimated	depth 2 (43)	0.58	0.37	0.07	-0.01	0.06	-0.06
estimated	depth $\geq$ 3 (82)	0.70	0.54	0.12	-0.10	0.10	-0.05
estimated, stabilised	downstream depth 1 (95)	0.36	0.30	0.29	0.30	0.30	0.30
estimated, stabilised	depth 2 (43)	0.58	0.55	0.51	0.51	0.52	0.52
estimated, stabilised	depth $\geq$ 3 (82)	0.70	0.71	0.67	0.66	0.68	0.68

Table S9: Truncated propagation on the four held-out knockdowns of the original design (seeds 0–9). [a]Algebraic inverse; equals the infinite sum only if $\rho(J) < 1$.

Model	D=1	2	3	4	6	8	12	$(I-J)^{-1}$ [a]	$\rho(J)$
no prior	0.50	0.27	0.09	0.00	0.03	-0.01	-0.01	0.15	3.03
hard identity mask + soft site prior	0.41	0.21	0.14	0.08	0.06	0.00	-0.01	0.18	2.53
no prior, stabilised (ρ=0.8)	0.50	0.45	0.43	0.45	0.44	0.44	0.44	0.44	0.80
hard identity mask + soft site prior, stabilised (ρ=0.8)	0.41	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.80

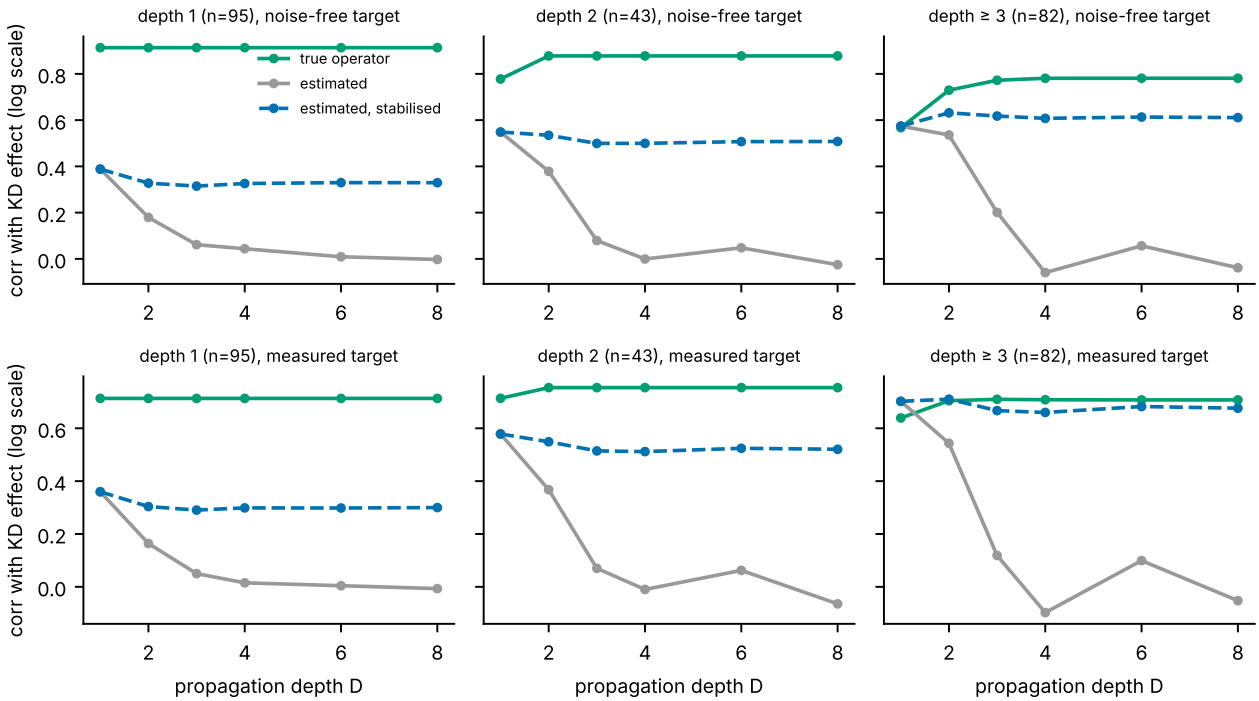


Figure S4: Propagation depth with the true vs estimated operator, both scored against the same noise-free (top) and measured (bottom) log-scale targets, stratified by each regulator's true downstream depth.

Table S10: Sign flips among strong shared targets (conditional analysis). Gain slope: total-least-squares slope of context-2 on context-1 effects.

Comparison	n	sign flips / strong shared	Jaccard	corr top-50	gain slope (median)	fold gain change
Norman 2019 (K562, CRISPRa; from earlier analysis)						
A alone vs A in C's context	260	– (1.5%)	0.185	–	–	–
A alone, split-half null	251	– (0.0%)	0.430	–	–	–
Cell-cycle state bins	152	5/10223 (0.05%)	–	0.96	–	1.26
Wessels 2023 (THP-1, Cas13 knockdown)						
A alone vs A in C's context	106	18/1657 (1.09%)	0.106	0.44	0.99	1.20
A alone, split-half null	106	1/1463 (0.07%)	0.110	0.85	1.00	1.12
Shifrut 2018 (primary T cells, CRISPR KO)						
No stimulation vs stimulation	18	0/6 (0.00%)	0.011	0.02	–	–
split-half null (no stim.)	18	0/6 (0.00%)	0.010	-0.02	–	–
split-half null (stim.)	18	0/12 (0.00%)	0.020	0.12	–	–

Table S11: Noise-constant estimators. Pooled and median per-knockdown expected-noise shares under alternative estimators of the noise constant, and a bootstrap over the 585 non-targeting controls (median estimator). Per-gene medians of squared z-scores are biased low for chi-square-like distributions and are shown divided by the χ^2_1 median (0.455).

Noise-constant estimator	pooled noise share	median per-knockdown noise share
median over controls (used)	0.470	0.737
mean over controls	0.499	0.782
20% trimmed mean	0.471	0.738
gene-specific mean	0.499	0.782
gene-specific median / chi2 median (0.455)	0.475	0.745
Bootstrap over 585 controls (95% interval, median estimator)	[0.467, 0.473]	[0.733, 0.742]

Table S12: Matched shared axes across screens ($|\cos|$ between individual axes fitted separately in each screen on the 1,457 knockdowns in both).

GW axis \ essential axis	1	2	3
1	0.95	0.17	0.08
2	0.16	0.91	0.13
3	0.05	0.12	0.02

Table S13: Positive-control robustness. Mean difference between cascade and random-mediator correlations for semi-synthetic targets with exact cascade share α (realised share equals nominal in orthogonalised generators; in the non-orthogonal generator the realised share was 0.005, 0.053, 0.102, 0.250 and 0.499). *: 95% CI lower bound above the real-data upper bound (0.004).

Generator	coverage	cascade share α (exact energy fraction)				
		0	0.05	0.1	0.25	0.5
base (orthogonal, strongest mediators, coordinates restored, equal reliability)	1.0	0.000	0.047*	0.066*	0.106*	0.153*
base (orthogonal, strongest mediators, coordinates restored, equal reliability)	0.25	-0.002	0.008*	0.013*	0.026*	0.039*
mediators ranked at random (ranking errors)	1.0	-0.003	0.006	0.009*	0.017*	0.023*
mediators ranked at random (ranking errors)	0.25	-0.003	-0.001	-0.001	0.003	0.005
mediator coordinates not restored	1.0	-0.004	-0.002	-0.002	-0.000	0.001
mediator coordinates not restored	0.25	0.000	-0.001	-0.002	-0.003	0.001
heterogeneous mediator strengths (lognormal, sd 1)	1.0	-0.003	0.035*	0.049*	0.080*	0.115*
heterogeneous mediator strengths (lognormal, sd 1)	0.25	-0.004	0.003	0.008*	0.015*	0.027*
screen-2 reliability halved	1.0	0.000	0.033*	0.046*	0.074*	0.107*
screen-2 reliability halved	0.25	-0.001	0.005	0.009*	0.018*	0.027*
screen-1 reliability halved	1.0	-0.003	0.032*	0.047*	0.074*	0.109*
screen-1 reliability halved	0.25	-0.003	0.006	0.009*	0.018*	0.026*
non-orthogonal construction (original)	1.0	-0.002	0.044*	0.064*	0.104*	0.152*
non-orthogonal construction (original)	0.25	-0.003	0.008*	0.013*	0.025*	0.038*
all adverse: random ranking, no restore, heterogeneous, screen-2 halved	1.0	-0.001	-0.003	-0.003	-0.001	-0.000
all adverse: random ranking, no restore, heterogeneous, screen-2 halved	0.25	-0.003	-0.004	-0.004	-0.001	-0.001
Real data (specific residual)	–	-0.004 [-0.012, 0.004]				

Table S14: Calibrated magnitude scores on independent-screen targets (R_0^2 uncalibrated; after training-only scalar calibration with bootstrap 95% CI; mean calibration factor $\bar{\lambda}$ over folds).

	Indep., raw			Indep., mean-centred			Indep., specific		
Predictor	R_0^2	R_0^2 calibrated [95% CI]	$\bar{\lambda}$	R_0^2	calibrated	$\bar{\lambda}$	R_0^2	calibrated	$\bar{\lambda}$
mean training profile	0.007	0.033 [0.029, 0.036]	0.53	0.000	0.000 [0.000, 0.000]	0.00	-0.000	-0.000 [-0.000, 0.000]	8.00
kNN k=10	-0.189	0.037 [0.032, 0.042]	0.28	-0.174	0.006 [0.004, 0.009]	0.15	-0.120	0.002 [0.001, 0.003]	0.10
kNN k=20	-0.117	0.042 [0.036, 0.048]	0.35	-0.106	0.007 [0.003, 0.010]	0.20	-0.056	0.003 [0.002, 0.003]	0.16
kNN k=50	-0.064	0.046 [0.040, 0.052]	0.40	-0.056	0.006 [0.002, 0.010]	0.26	-0.017	0.004 [0.003, 0.005]	0.27
kNN k=100	-0.039	0.047 [0.041, 0.052]	0.42	-0.033	0.005 [0.002, 0.008]	0.28	-0.004	0.004 [0.003, 0.005]	0.36
factorised (generic mean + pathway kNN50 + specific kNN50)	-0.017	0.043 [0.037, 0.049]	0.46	-0.016	0.012 [0.009, 0.016]	0.40	-0.017	0.004 [0.003, 0.005]	0.27

Table S15: Paired predictor comparisons (per-knockdown differences; bootstrap 95% CI).

Paired comparison	target	Δ corr [95% CI]	ΔR_0^2 calibrated [95% CI]
kNN k=100 minus kNN k=10	raw	0.041 [0.034, 0.048]	0.0097 [0.0064, 0.0127]
kNN k=100 minus kNN k=10	mean-centred	0.020 [0.009, 0.030]	-0.0011 [-0.0034, 0.0010]
kNN k=100 minus kNN k=10	specific residual	0.023 [0.016, 0.030]	0.0024 [0.0016, 0.0031]
kNN k=50 minus kNN k=10	raw	0.040 [0.033, 0.046]	0.0089 [0.0056, 0.0118]
kNN k=50 minus kNN k=10	mean-centred	0.022 [0.013, 0.031]	-0.0001 [-0.0024, 0.0020]
kNN k=50 minus kNN k=10	specific residual	0.016 [0.010, 0.022]	0.0018 [0.0011, 0.0024]
kNN k=100 minus kNN k=50	raw	0.002 [-0.001, 0.004]	0.0008 [-0.0005, 0.0022]
kNN k=100 minus kNN k=50	mean-centred	-0.002 [-0.007, 0.003]	-0.0010 [-0.0019, 0.0000]
kNN k=100 minus kNN k=50	specific residual	0.007 [0.002, 0.011]	0.0006 [0.0002, 0.0010]
factorised (generic mean + pathway kNN50 + specific kNN50) minus kNN k=50	raw	-0.007 [-0.009, -0.005]	-0.0031 [-0.0048, -0.0012]
factorised (generic mean + pathway kNN50 + specific kNN50) minus kNN k=50	mean-centred	0.005 [-0.002, 0.012]	0.0064 [0.0046, 0.0083]
factorised (generic mean + pathway kNN50 + specific kNN50) minus kNN k=50	specific residual	0.000 [-0.000, 0.000]	-0.0000 [-0.0000, 0.0000]